Yasuhide Mochizuki, Ph.D.

Junior Associate Professor at *Tokyo University of Science*

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Research map Web of Science

Employment History

2026/4 – Present **Junior Associate Professor,** Department of Materials Science and Technology, Faculty of Advanced Engineering,
Tokyo University of Science.

2024/10 – 2026/3 **Assistant Professor,** Department of Materials Science and Engineering, School of Materials and Chemical Technology,
Institute of Science Tokyo (formerly known as Tokyo Institute of Technology).

2025/7 – 2026/3 **Visiting Researcher,** Materials DX Research Center,
National Institute of Advanced Industrial Science and Technology (AIST).

2021/4 – 2024/9 **Assistant Professor,** Department of Materials Science and Engineering, School of Materials and Chemical Technology,
Tokyo Institute of Technology.

2018/4 – 2021/3 **JSPS Research Fellow,** Laboratory for Materials and Structures,
Tokyo Institute of Technology.

Education

2018/4 – 2021/3 **Ph. D. in Engineering, Tokyo Institute of Technology**
Department of Materials Science and Engineering.
Dissertation title:
First-principles lattice-dynamics study of structural distortions in perovskites.
Supervisor: Prof. Fumiyasu Oba

2016/4 – 2018/3 **Master in Engineering, Tokyo Institute of Technology**
Department of Materials Science and Engineering.
Thesis title: *First-principles lattice-dynamics study of strain-induced structural transitions in layered perovskites.*
Supervisor: Prof. Fumiyasu Oba

2012/4 – 2016/3 **Bachelor in Engineering, Tokyo Institute of Technology**
Department of Inorganic Materials.

2009/4 – 2012/3 **Shizuoka Kita High School**
Science and Mathematics Course.

Research Publications (First & Corresponding Authors)

1. H. Koiso, S. Yoshida, T. Nagai, T. Isobe, A. Nakajima, and **Y. Mochizuki***,
"Reply to Comment on "Thermal expansion and phase stability of BF_3 (B = Sc, Y, La, Al, Ga, In) from first principles"",
Physical Review B 112, 226102 (2025).

2. H. Koiso, S. Kawasoe, J. Zhang, T. Isobe, A. Nakajima, and **Y. Mochizuki***,
"Discovery of Biaxial Negative Thermal Expansion in $KNbSi_2O_7$ from Computations and Experiments",
The Journal of Physical Chemistry C 129, 9509-9517 (2025).

3. R. Kiribayashi, K. Sunada, T. Isobe, K. Kobayashi, T. Nagai, H. Ishiguro, **Y. Mochizuki***, and A. Nakajima*,
"Preparation and Antiviral Activity of La_2CuO_4 and $Y_2Cu_2O_5$ with Mechanistic Insights from First Principles",
ACS Applied Materials & Interfaces 17, 61872-61887 (2025).

4. H. Koiso, S. Yoshida, T. Nagai, T. Isobe, A. Nakajima, and **Y. Mochizuki***,
"Thermal expansion and phase stability of BF_3 (B = Sc, Y, La, Al, Ga, In) from first principles",
Physical Review B 110, 064104-1-12 (2024).

5. **Y. Mochizuki***,†, K. Nagamatsu†, H. Koiso, T. Isobe, and A. Nakajima,
"Mechanism of Negative Thermal Expansion in Monoclinic $Cu_2P_2O_7$ from First Principles",
The Journal of Physical Chemistry Letters 15, 156-164 (2024). †Equal contribution

6. **Y. Mochizuki***, H. Koiso, K. Nagamatsu, S. Bae, T. Isobe, and A. Nakajima,
"Thermal Properties of the Element and Binary Oxides toward Negative Thermal Expansion: A First-Principles Lattice-Dynamics Study",
The Journal of Physical Chemistry C 128, 525-535 (2024).

7. **Y. Mochizuki***, H.-J. Sung, T. Gake, and F. Oba*,
"Chemical Trends of Surface Reconstruction and Band Positions of Nonmetallic Perovskite Oxides from First Principles",
Chemistry of Materials 35, 2047-2457 (2023).

8. **Y. Mochizuki**, T. Nagai, H. Shirakuni, A. Nakano, F. Oba, I. Terasaki, and H. Taniguchi*,
"Coexisting Mechanisms for Ferroelectric Phase Transition in $Li_2SrNb_2O_7$",
Chemistry of Materials 33, 1257-1264 (2021).

9. **Y. Mochizuki***, H.-J. Sung, A. Takahashi, Y. Kumagai, and F. Oba,
"Theoretical exploration of mixed-anion antiperovskite semiconductors M_3XN (M = Mg, Ca, Sr, Ba; X = P, As, Sb, Bi)",
Physical Review Materials 4, 044601-1-14 (2020).

10. T. Nagai†, **Y. Mochizuki**†, H. Shirakuni, A. Nakano, F. Oba, I. Terasaki, and H. Taniguchi*,
"Phase Transition from Weak Ferroelectricity to Incipient Ferroelectricity in $Li_2Sr(Nb_{1-x}Ta_x)_2O_7$",
Chemistry of Materials 32, 744-750 (2020). †Equal contribution.

11. **Y. Mochizuki***, Y. Kumagai, H. Akamatsu, and F. Oba,
"Polar metallic behavior of strained antiperovskites $ACNi_3$ (A = Mg, Zn, and Cd) from first principles",
Physical Review Materials 2, 125004-1-10 (2018).

12. **Y. Mochizuki**, H. Akamatsu*, Y. Kumagai, and F. Oba,
"Strain-engineered Peierls instability in layered perovskite $La_3Ni_2O_7$ from first principles",
Physical Review Materials 2, 125001-1-7 (2018).

Other Research Publications

13. T. Nagai*, **Y. Mochizuki**, S. Yoshida, and T. Kimura*,
"Chemical Aspect of Displacive-Type Ferroaxial Phase Transition from Perspective of Second-Order Jahn-Teller Effect: NASICON Systems as an Example",
Journal of the American Chemical Society 145, 8090-8098 (2023).

14. H.-J. Sung*, **Y. Mochizuki**, and F. Oba*,
"Surface reconstruction and band alignment of nonmetallic A(II)B(IV)O_3 perovskites",
Physical Review Materials 4, 044606-1-13 (2020).

15. M. Hemmi, N. Arimitsu, **Y. Mochizuki**, A. Nakajima, and T. Isobe*,

"Water desorption-driven negative thermal expansion behavior of synthesized potassium-ion-exchanged gismondine zeolite",
Materials Chemistry and Physics 350, 131936 (2026).

16. B. Zhang, D. Moriyama, T. M. Herath, H. Zhao, Y. Bai, Y. Dang, H. Kato, K. Yamamoto, **Y. Mochizuki**, N. Arimitsu, A. Nakajima, X. Zhu, B. Chen, M. Fujii*, and T. Isobe*,
"Surface engineering boosted adsorption of PFAS on porous carbon: Experimental and theoretical investigations",
Journal of Hazardous Materials 499, 140063 (2025).

17. M. Khazaei*, M. Bagheri, A. Ranjbar*, S. Bae*, R. Khaledialidusti, **Y. Mochizuki**, T. D. Kühne, K. Shudo, H. Raebiger, H.-P. Komsa*, and H. Hosono,
"Diverse Surface Reconstructions in MAX Phases",
Nanoscale 17, 24184-24195 (2025).

18. F. Oba*, T. Nagai, R. Katsube, **Y. Mochizuki**, M. Tsuji, G. Deffrennes, K. Hanzawa, A. Nakano, A. Takahashi, K. Terayama, R. Tamura, H. Hiramatsu, Y. Nose, and H. Taniguchi,
"Theoretical and data-driven approaches to semiconductors and dielectrics: from prediction to experiment",
Science and Technology of Advanced Materials 25, 2423600-1-33 (2024).

19. M. Khazaei*, S. Bae*, R. Khaledialidusti, A. Ranjbar, H.-P. Komsa, S. Khazaei, M. Bagheri, V. Wang, **Y. Mochizuki**, M. Kawamura, G. Cuniberti, S. M. Allaei, K. Ohno, H. Hosono, and H. Raebiger*,
"Superlattice MAX phases with A-layers reconstructed into 0D-clusters, 1D-chains, and 2D-lattices",
The Journal of Physical Chemistry C 127, 14906-14913 (2023).

20. A. Nakano*, H. Shirakuni, T. Nagai, **Y. Mochizuki**, F. Oba, H. Yokota, S. Kawaguchi, I. Terasaki, and H. Taniguchi,
"Phase variation of ferroelectric $Li_2Sr_{1-x}Ca_x(Nb_{1-x}Ta_x)_2O_7$00Aoby selective reinforcement in the (Nb, Ta)-O covalent bonds",
Physical Review Materials 6, 044412-1-7 (2022).

21. A. Takahashi*, Y. Kumagai, J. Miyamoto, **Y. Mochizuki**, and F. Oba,
"Machine learning models for predicting the dielectric constants of oxides based on high-throughput first-principles calculations",
Physical Review Materials 4, 103801-1-13 (2020).

22. N. Matsunaga*, S. Imai, T. Shirokura, **Y. Mochizuki**, K. Kuwahata, K. Tsutsui, K. Kakushima, and H. Wakabayashi,
"PVD-MoS_2 film crystallinity deposited on underlying insulators discussing with molybdenum antisites",
IEEE Journal of the Electron Device Society 13, 883-891 (2025).

23. R. Kiribayashi, K. Sunada, **Y. Mochizuki**, T. Isobe, K. Kobayashi, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Preparation of $La_2Sn_2O_7$, Mg_2SnO_4, and $MgSn(OH)_6$ and their antiviral and antibacterial activities",
Journal of Materials Science: Materials in Medicine 36, 64-1-14 (2025).

24. N. Ogawa, **Y. Mochizuki**, T. Isobe, K. Kobayashi, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Processing and antiviral activity of Cu-modified $(Ce_{0.8}, Gd_{0.2})O_{2-\delta}$ in the dark and under visible light", *Journal of the Ceramic Society of Japan* 133, 474-480 (2025).

25. K. Abe, K. Sunada, **Y. Mochizuki**, T. Isobe, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Effects of UV illumination on organic dye decomposition activity and antibacterial and antiviral activities of rare earth iodates",
Journal of Materials Science 59, 17558-17572 (2024).

26. K. Abe, K. Sunada, **Y. Mochizuki**, T. Isobe, T. Nagai, H. Ishiguro, and A. Nakajima*,

"Antibacterial and antiviral activities of transparent PVA coating films prepared by using solutions containing eluted ions from rare earth iodates",
Journal of Coatings Technology and Research 22,471-480 (2025).

27. R. Kiribayashi, R. Nakane, K. Sunada, **Y. Mochizuki**, T. Isobe, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Antiviral and antibacterial activities of yttria-stabilized zirconia",
Materials Letters 367, 136658-1-3 (2024).

28. R. Nakane, R. Kiribayashi, K. Sunada, **Y. Mochizuki**, T. Isobe, S. Matsushita, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Degradation of 2-naphthol in water and antibacterial/antiviral activity by Zn_2SnO_4, $ZnSn(OH)_6$ and $Y_2Sn_2O_7$",
Ceramics International 50, 10797-10805 (2024).

29. N. Kobayashi, K. Sunada, **Y. Mochizuki**, T. Isobe, S. Matsushita, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Decomposition of 2-naphthol in water and antiviral activity of CoO_x modified ($Ce_{0.8}$, $Bi_{0.2-y}$, La_y) $O_{2-\delta}$ in the dark",
Journal of the Ceramic Society of Japan 131, 590-594 (2023).

30. T. Sakai, K. Sunada, **Y. Mochizuki**, T. Isobe, S. Matsushita, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Antiviral and antifungal activities of tin molybdenum solid solution oxide prepared using mechanochemical processing",
Journal of the Ceramic Society of Japan 131, 482-487 (2023).

31. R. Kiribayashi, K. Sunada, **Y. Mochizuki**, T. Isobe, S. Matsushita, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Decomposition of 2-naphthol in water and its antibacterial and antiviral activities by $LaMnO_3$ and $LaCoO_3$ in the dark",
Journal of the Ceramic Society of Japan 131, 117-125 (2023).

32. S. Iwakura, T. Ito, T. Sakai, K. Sunada, **Y. Mochizuki**, T. Isobe, S. Matsushita, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Processing of cerium molybdates by solid phase process and their antiviral activity",
Journal of the Japan Society of Colour Material 96, 96-103 (2023).

33. K. Abe, K. Sunada, **Y. Mochizuki**, T. Isobe, S. Matsushita, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Preparation of rare earth iodates and their decomposition activity on organic dyes and antibacterial/antiviral activities",
Ceramics International 49, 14681-14688 (2023).

34. T. Hirosawa, K. Yamamoto, M. Sakai, **Y. Mochizuki**, T. Isobe, S. Matsushita, and A. Nakajima*,
"Time dependence of the dominant mechanisms of self-propelled droplets by Leidenfrost phenomenon on Zn plate surfaces with and without ZnO nanorods",
Materials Chemistry and Physics 288, 126349-1-9 (2022).

35. T. Sakai, K. Sunada, **Y. Mochizuki**, T. Isobe, S. Matsushita, T. Nagai, H. Ishiguro, and A. Nakajima*,
"Antiviral and antifungal activities of lanthanum molybdate and copper molybdate",
Journal of the Ceramic Society of Japan 130, 370-375 (2022).

36. Y. Nakamura, N. Arimitsu, **Y. Mochizuki**, S. Matsushita, A. Nakajima, and T. Isobe*,
"Synthesis of Amino-Functionalized Amorphous Carbon Membranes via Glucose-solution Hydrothermal Method",
Journal of the Ceramic Society of Japan, in press.

37. D. Nagai, H. Igawa, N. Arimitsu, **Y. Mochizuki**, A. Nakajima, T. Isobe*,
"Preparation and Thermal Expansion Behavior of an EDI-Type SAPO with Dehydration-Induced Negative Thermal Expansion",

Journal of the Ceramic Society of Japan, in press.

Invited Presentations

1. **Y. Mochizuki***, H. Koiso, K. Nagamatsu, S. Bae, T. Isobe, and A. Nakajima,
"Comprehensive First-Principles Lattice Dynamics Study for Thermal Properties of the Element and Binary Oxides toward Negative Thermal Expansion",
The 39th International Korea-Japan Seminar on Ceramics (KJ-Ceramics 39), Jeju, Korea, Nov. 2025.

2. **Y. Mochizuki***,
"Mechanism Elucidation of Phonon-Induced Negative Thermal Expansion from First Principles",
33rd Annual Meeting of MRS-J, Yokohama, Japan, Nov. 2023.

3. **Y. Mochizuki***,
"Structural exploration of perovskites using first-principles lattice-dynamics calculations",
30th Annual Meeting of MRS-J, Online, Japan, Dec. 2020.

Awards

2023/8 **Tokyo Tech Challenging Research Award**.

2023/1 **Award for Encouragement of Research, the 32nd Annual Meeting of MRS-J Symposium**.

2022/9 **IUMRS-ICYRAM 2022, Young Researcher Award (Silver Prize)**.

2022/3 **Seiichi Tejima Research Award (Doctoral Dissertation Award)**.

2018/12 **Award for Encouragement of Research, the 28th Annual Meeting of MRS-J Symposium**.

2018/1 **Good Presentation Award, World Young Fellow Meeting 2018, the 56th Symposium of Basic Science of Ceramics**.

Grants & Fundings

1. **KAKENHI Grant-in-Aid for Scientific Research (B)**
Project period: Apr. 2025 - Mar. 2029
Total budget: ¥18,460,000

2. **AIST Kakusei Project**
Project period: Jul. 2025 - Mar. 2026
Total budget: ¥3,000,000

3. **The Kazuchika Okura Memorial Foundation**
Project period: Dec. 2023 - Mar. 2025
Total budget: ¥1,500,000

4. **Nippon Sheet Glass Foundation for Materials Science and Engineering**
Project period: Apr. 2022 - Mar. 2025
Total budget: ¥1,200,000

5. **Tokyo Tech Challenging Research Award**
Project period: Jul. 2023 - Mar. 2024
Total budget: ¥2,600,000

6. **IKETANI Science and Technology Foundation**
Project period: Apr. 2022 - Mar. 2023

Total budget: ¥1,000,000

7. **KAKENHI Grant-in-Aid for Early-Career Scientists**
   Project Period: Apr. 2022 - Mar. 2025
   Total Budget: ¥4,030,000

8. **The Kurata Grants of the Hitachi Global Foundation**
   Project period: Jan. 2022 - Mar. 2023
   Total budget: ¥1,000,000

9. **KAKENHI Grant-in-Aid for JSPS Fellows**
   Project period: Apr. 2018 - Mar. 2021
   Total budget: ¥2,200,000